Cervical Cytology-Histology Correlation QA Report

Automated rule-based quality assurance analysis

Total Cases: 31

Executive Summary

A total of 31 cases were analyzed using a deterministic rule-based cervical cytology-histology correlation quality assurance pipeline. Concordance, discordance subtypes, HSIL-focused metrics, and structured visual outputs are included in this report.

Concordance Summary

Concordance_Class	Count	Percent
minor_discordant	14	45.16
major_discordant	9	29.03
concordant	5	16.13
unmapped	3	9.68

HSIL-Focused Metrics

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	0.0
HSIL Pap tests with minor discrepancies	66.67
HSIL Pap tests with major false-positive discrepancies	33.33
Number of major false-negative / cytology undercall discrepancies	7.0
PV+ for HSIL Pap tests (%)	0.0

Discrepancy Bucket Summary

Bucket	Count	Percent
MinVar	0	0.0
MajUnd	7	22.58
MinUnd	14	45.16
Agree	5	16.13
MinOver	0	0.0
MajOver	2	6.45

Key Quality Signals

- Major discordance rate: 29.03%

- Minor discordance rate: 45.16%

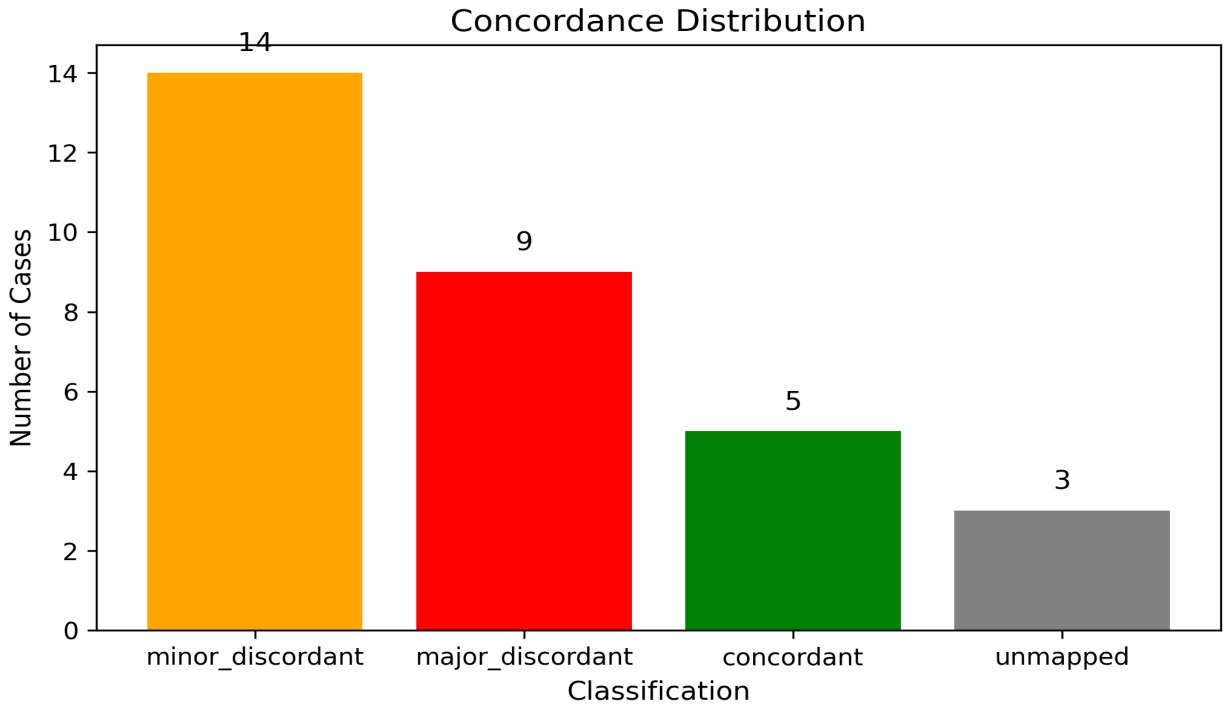
- Undercall proportion: 67.74%

- Overcall proportion: 6.45%

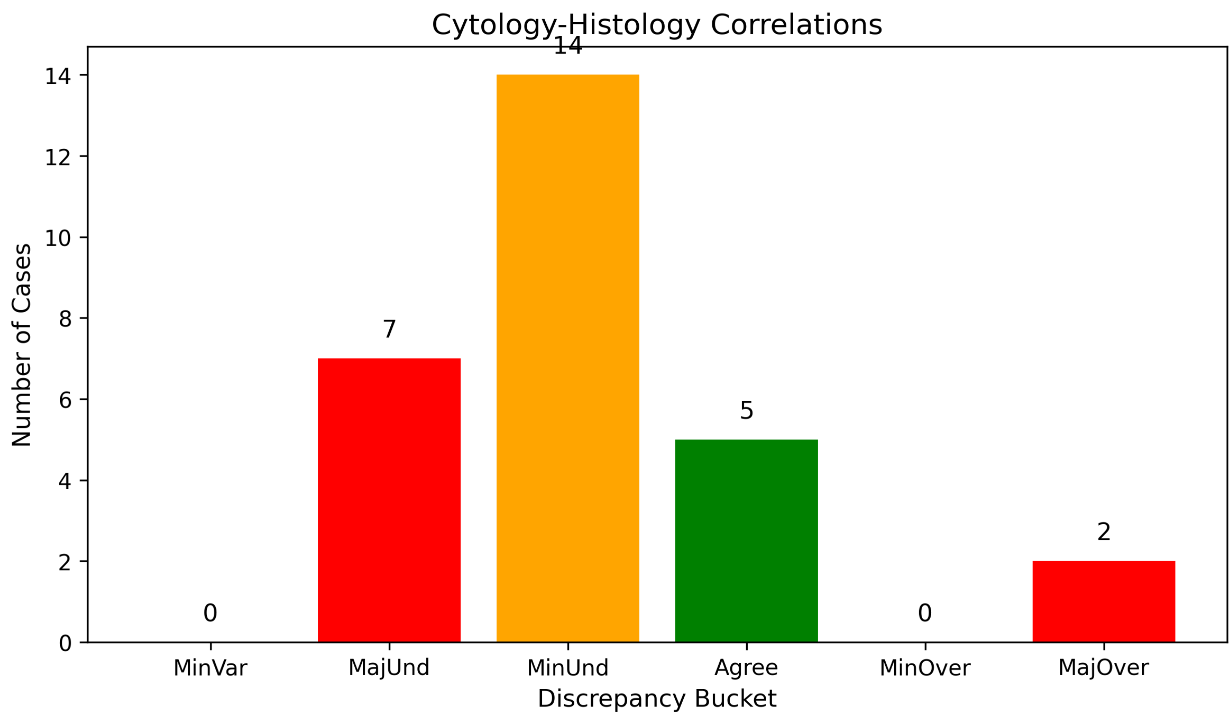
- HSIL PV+: 0.0%

Figures

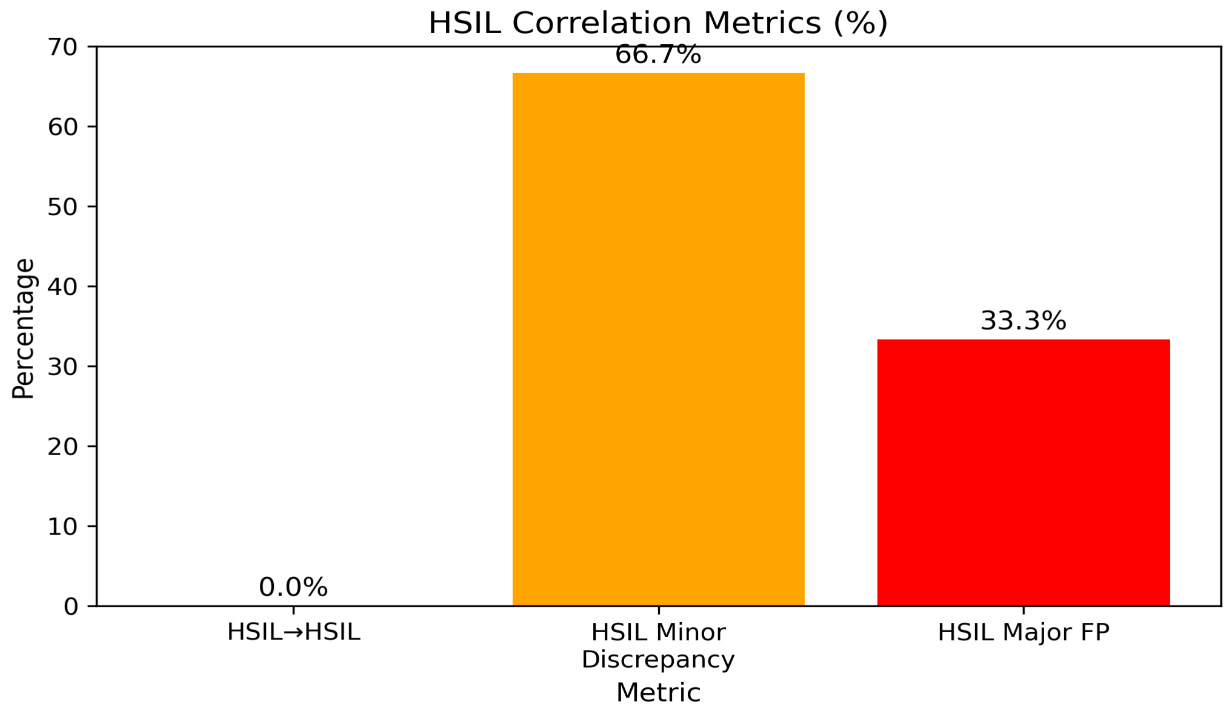
Concordance Distribution



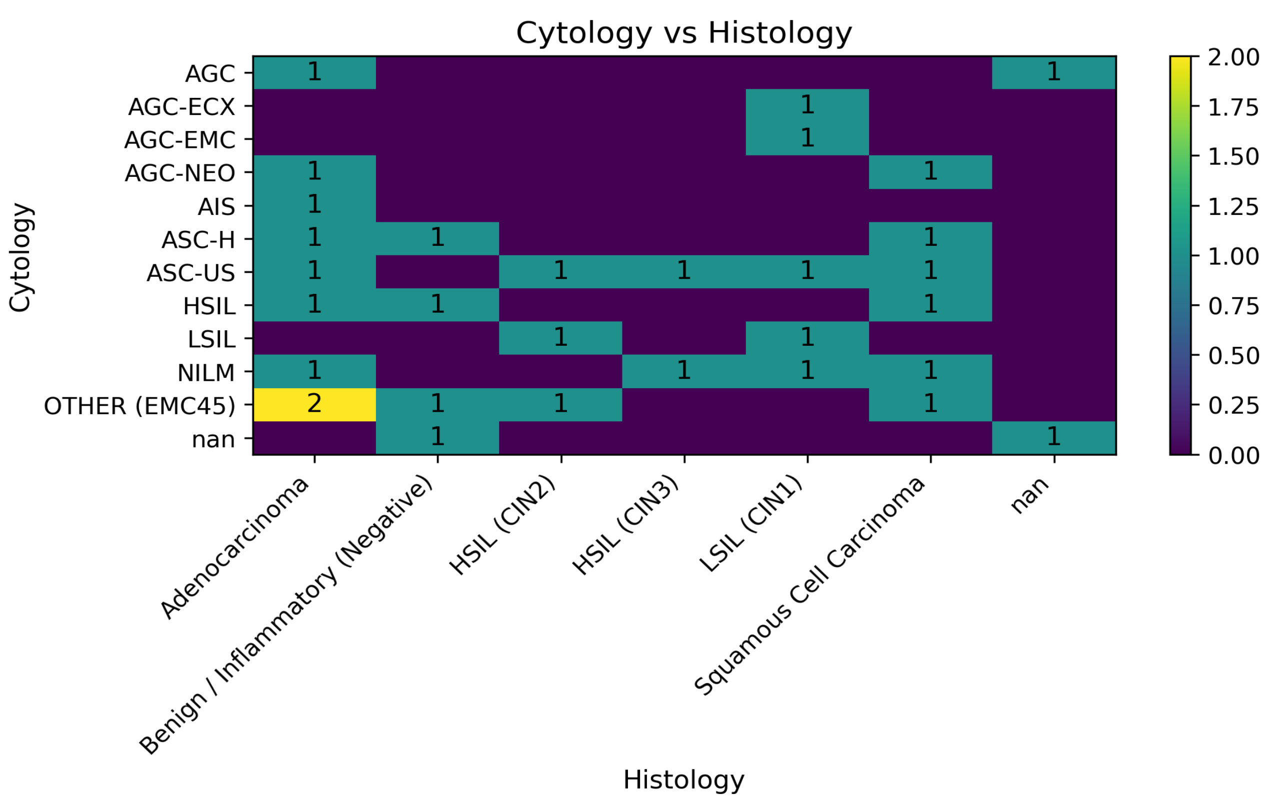
Cytology-Histology Correlation Buckets



HSIL Correlation Metrics



Confusion Matrix



Limitations

This report is based on deterministic cytology-histology correlation using rule-based classification.
Routine CHC datasets are subject to verification bias due to selective histologic follow-up.
Accordingly, these results should not be interpreted as measures of screening sensitivity or specificity.
Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.